Advanced Internet of Things Technology

Week 5: 6G Connectivity and Non-Terrestrial Networks

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Lecture Overview

Lecture positioning

- Advanced IoT Technology — 6G Connectivity and Non-Terrestrial Networks focus
- Standards-driven perspective: IMT-2030 framework + 3GPP Releases pathway

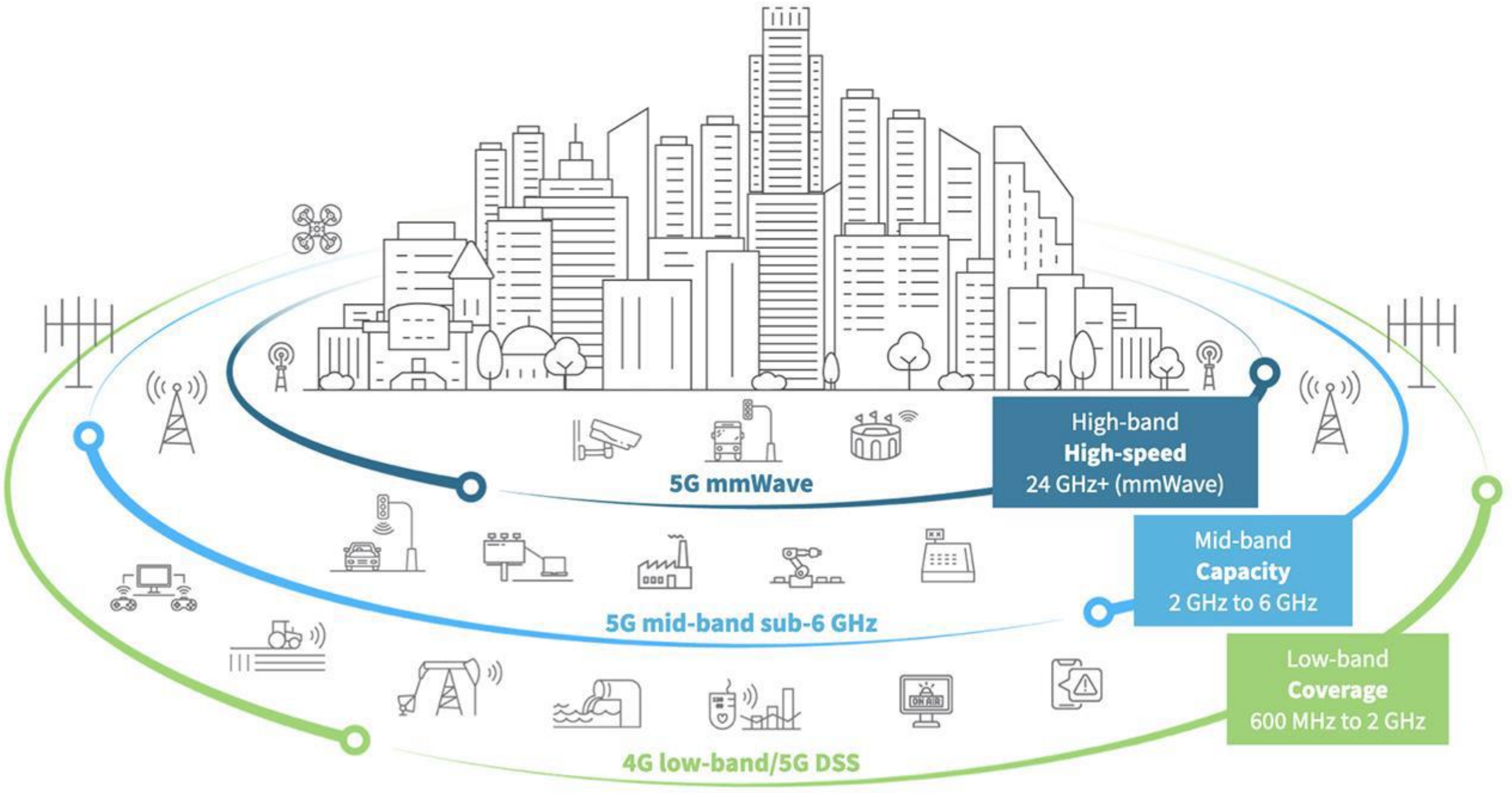
Why standards matter for next-generation IoT

- IoT evolution is now guided by formal global frameworks
- Capability targets defined by IMT-2030
- Implementation path defined by 3GPP Release evolution
- Long-range and NTN IoT depend strongly on standards support

Learning objectives

- Explain IMT-2030 capability dimensions and their IoT implications
- Describe how 3GPP Releases evolve IoT and NTN features
- Distinguish short-range vs long-range 6G IoT connectivity layers
- Analyse NTN-enabled IoT architecture patterns
- Evaluate deployment and interoperability trade-offs

5G IoT



Current 5G systems are mainly cellular networks

5G Expectations

Mission-critical services

Cellular Vehicle-to-Everything (C-V2X)

Drone communications | Private Networks

Ultra Reliable Low Latency Comms (URLLC)

Enhanced mobile broadband

Spectrum sharing | Flexible slot-based framework

Scalable OFDM | Massive MIMO | Mobile mmWave

Dual Connectivity | Advanced channel coding

Massive Internet of Things

Enhanced power save modes

Deeper coverage | Grant-free UL

Narrow bandwidth | Efficient signaling

10x Decrease in end-to-end latency

10x Experienced throughput

3x Spectrum efficiency

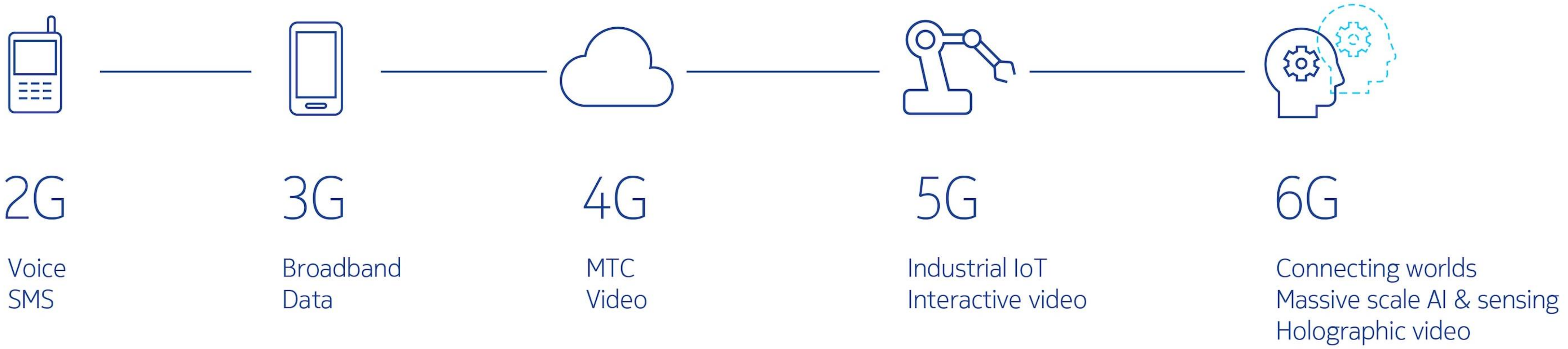
100x Traffic capacity

100x Network efficiency

10x Connection density

Motivation: From 5G IoT to 6G IoT

- Coverage gaps remain for remote and wide-area IoT
- Reliability and latency limits for critical IoT control loops
- Fragmented satellite and terrestrial IoT ecosystems
- Need unified terrestrial + non-terrestrial architecture



Comparative Analysis 5G vs. 6G

Feature	**5G IoT (NB-IoT/RedCap)**	**6G Ambient IoT (Release 19+)**	**6G NTN (Satellite)**
Power Source	Battery (Years)	Energy Harvesting (Infinite)	Solar/Battery (High Power)
Range	~10 km (Terrestrial)	< 100 m (Reader dependent)	Global (LEO/GEO)
Cost per Node	$5 - $20	< $0.10 - $1.00	$50+ (Terminal cost)
Spectrum	Sub-6 GHz / mmWave	UHF / n8 Guard Bands	Ka, Ku, S-Band
Latency	10ms - seconds	Variable (Harvesting dependent)	20ms (LEO) - 600ms (GEO)
Primary Use	Smart Metering, Wearables	Logistics, Retail, Inventory	Maritime, Rural, Transport

6G Vision Through IMT-2030 Framework

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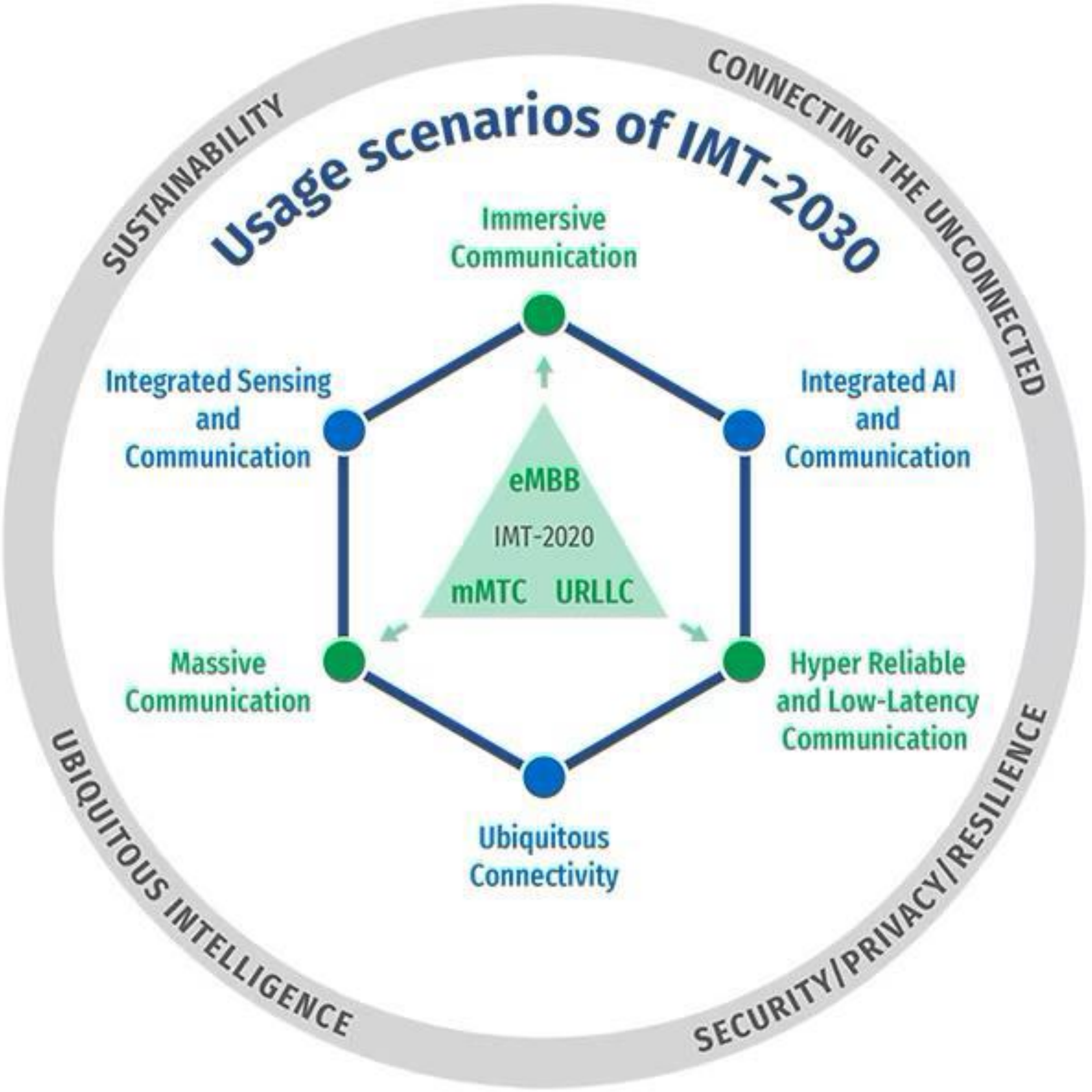
IMT evolution ladder

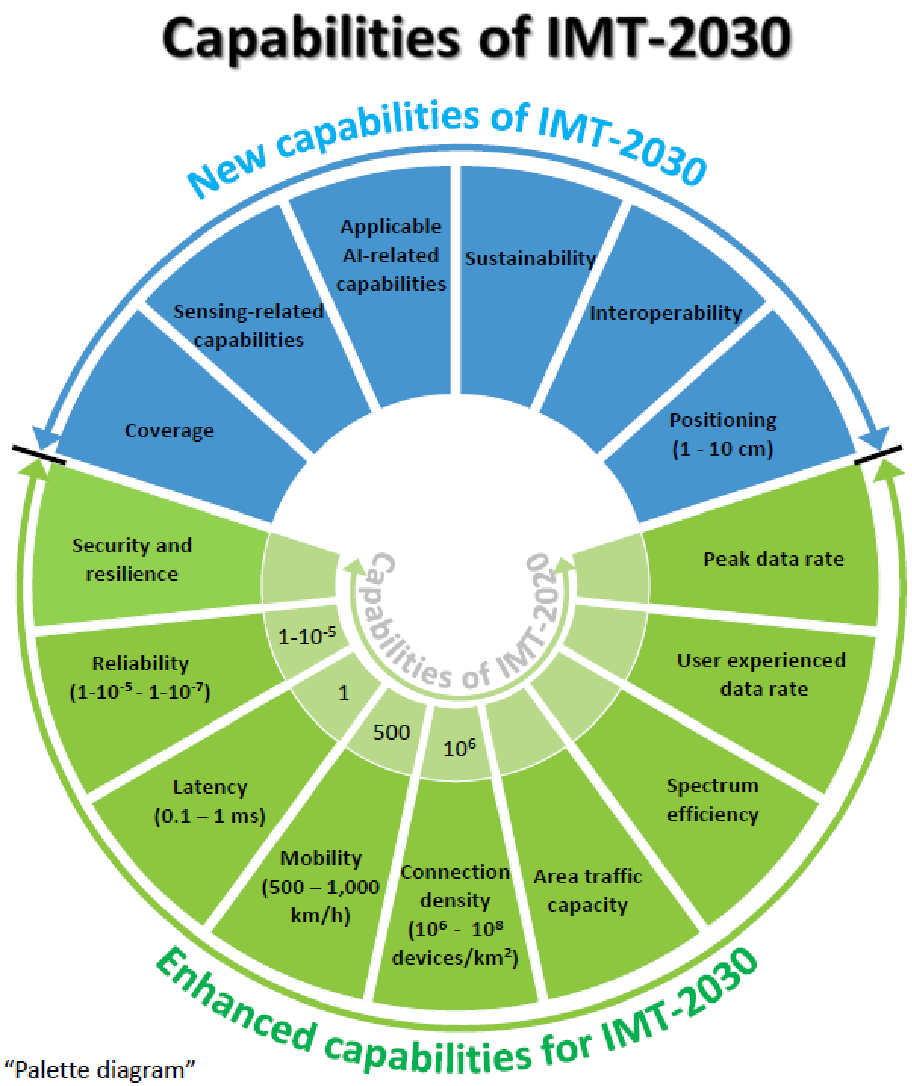
- IMT-2000 — 3G mobile systems
- IMT-Advanced — 4G broadband systems
- IMT-2020 — 5G multi-service systems
- IMT-2030 — 6G intelligent and integrated systems

Role of IMT-2030

- Defines capability targets for 6G systems
- Provides evaluation framework for candidate technologies
- Aligns global research and industry directions
- Expands focus beyond throughput to intelligence and sensing

6G Vision - ITU





So called "Palette diagram"

IMT-2030 usage scenario with IoT view

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IMT-2030 usage scenario families

- Massive ubiquitous connectivity — city and national IoT fabrics
- Hyper-reliable low latency — industrial and robotic IoT
- AI-native services — autonomous systems
- Integrated sensing — joint sensing-communication platforms
- Immersive communication — XR-assisted IoT operations

Insight

- IMT-2030 shifts IoT from device connectivity → system intelligence
- Networks become compute + sensing + control platforms

IMT-2030 capability dimensions

- IMT-2030 capability dimensions
  - Extreme connection density — massive sensor fabrics
  - Ultra-low latency — control and robotics IoT
  - Ultra-high reliability — safety and mission-critical IoT
  - Energy efficiency — battery and energy-harvesting IoT devices
  - AI-native networking — autonomous network control
  - Integrated sensing and communication — network as sensor
  - 3D global coverage — satellite and aerial IoT support
- IoT interpretation of capability dimensions
  - Density → smart city sensor grids
  - Latency → industrial automation loops
  - Reliability → remote infrastructure control
  - Energy → long-lifetime field sensors
  - AI → self-optimising IoT systems
  - Sensing → environment-aware networks
  - Coverage → rural and maritime IoT

6G Use Cases

Immersive multimedia beyond the living room

6G will bring VR, XR and cloud gaming beyond the indoors to any outdoor environment

Wireless connectivity everywhere, at any time

With NTN and other 6G network enhancements the access limitations of mobile networks will disappear, whether on a mountain, at sea or in the air

Robots automating our world

6G will connect and control the most advanced devices, including the robots that will assist us in our daily lives

A digital twin of reality

6G will be key to simulating massive environments, from factories to entire cities, unlocking unprecedented optimization capabilities

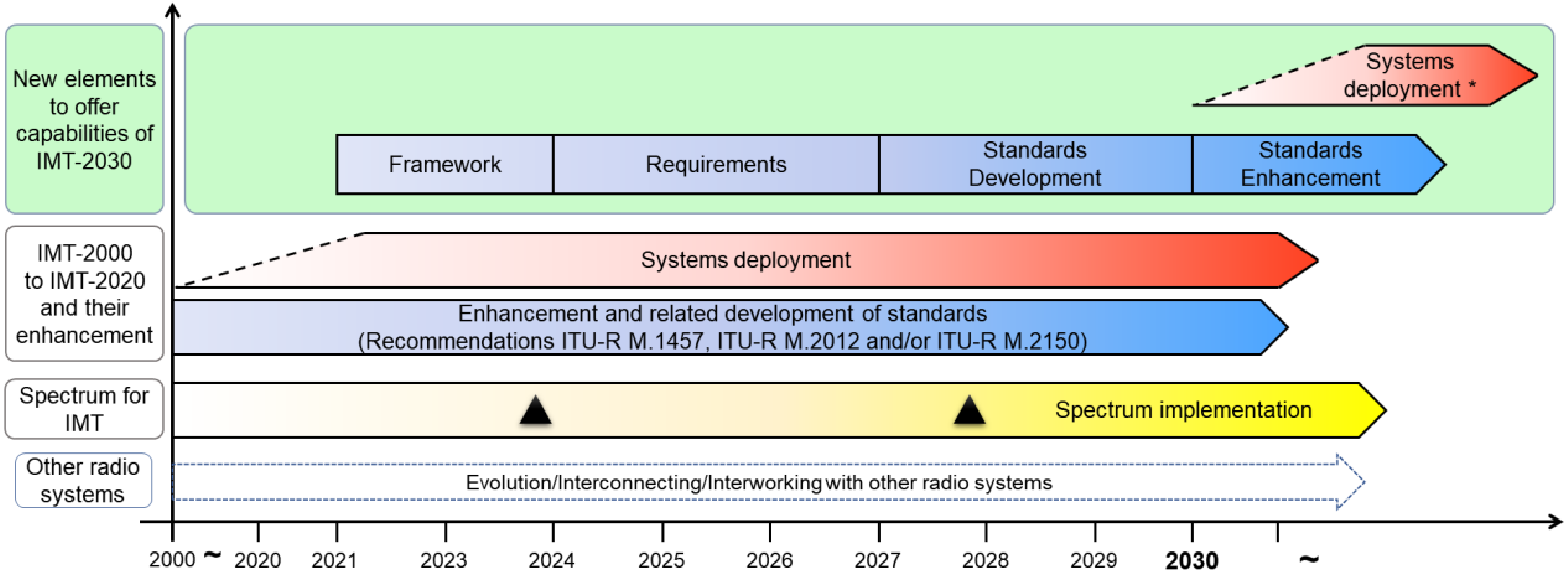
Swarms of autonomous AI agents saving lives

Drones, ground rovers and medical-triage software agents will work together in real time to assist in search-and-rescue operations

Digital-physical fusion

We will project life-sized, holographic representations of themselves into virtual or physical rooms, allowing genuine human interaction over any distance

Timelines for IMT-2030



The sloped dotted lines in systems deployment indicate that the exact starting point cannot yet be fixed.

▲ : Possible spectrum identification at WRC-23, WRC-27 and future WRCs

* : Systems to satisfy the technical performance requirements of IMT-2030 could be developed before year 2030 in some countries.
: Possible deployment around the year 2030 in some countries (including trial systems)

3GPP Release Evolution Toward 6G and NTN IoT

Why study 3GPP Releases

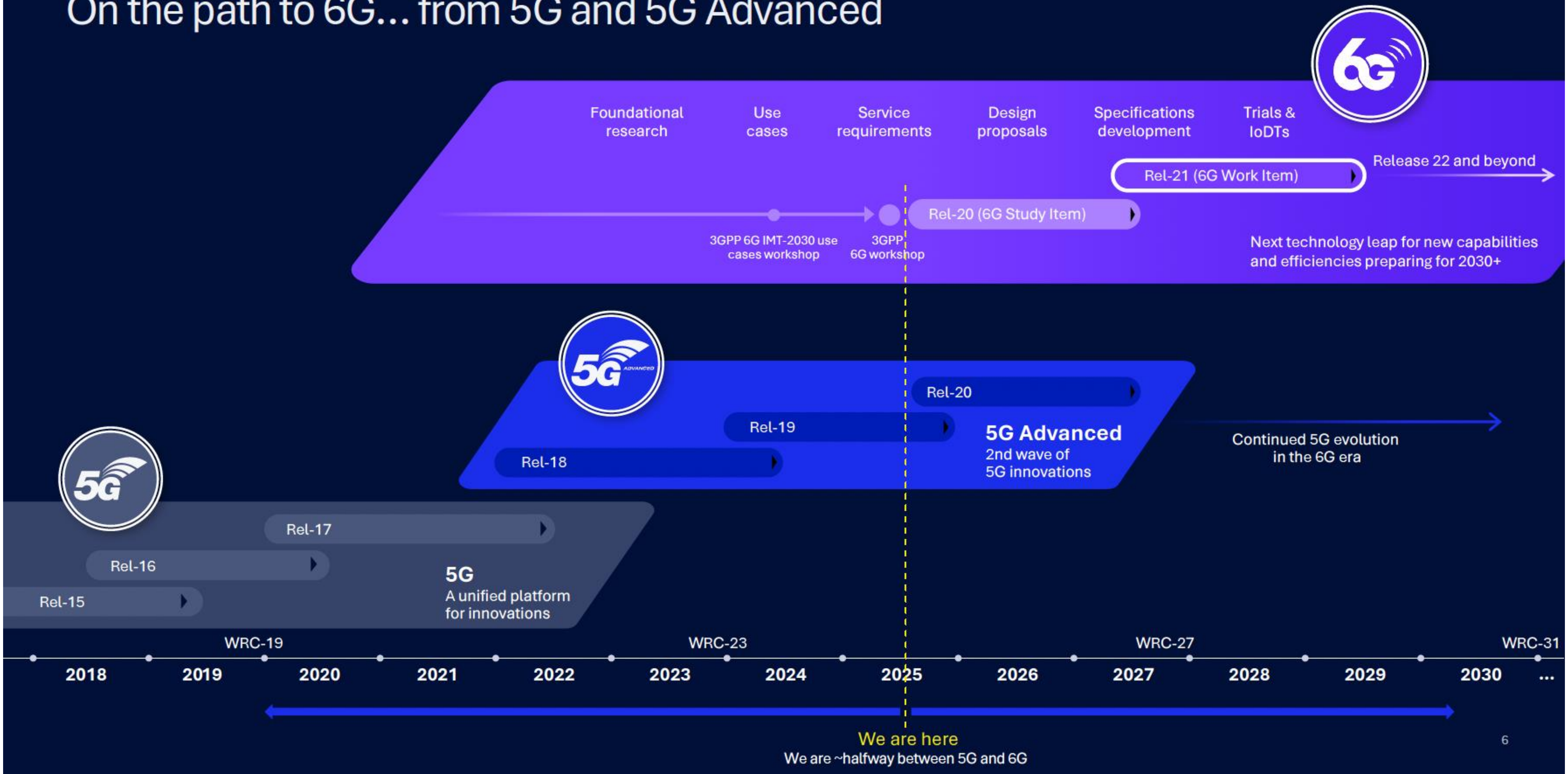
3GPP Releases in an IoT course

- Each Release introduces deployable capabilities
- Shows engineering pathway from concept to network feature
- Critical for understanding NTN and satellite IoT integration

Release structure concept

- Release = frozen feature set
- Devices and networks implement per Release level
- Later Releases extend earlier capabilities

On the path to 6G… from 5G and 5G Advanced



Releases relevant to IoT and NTN

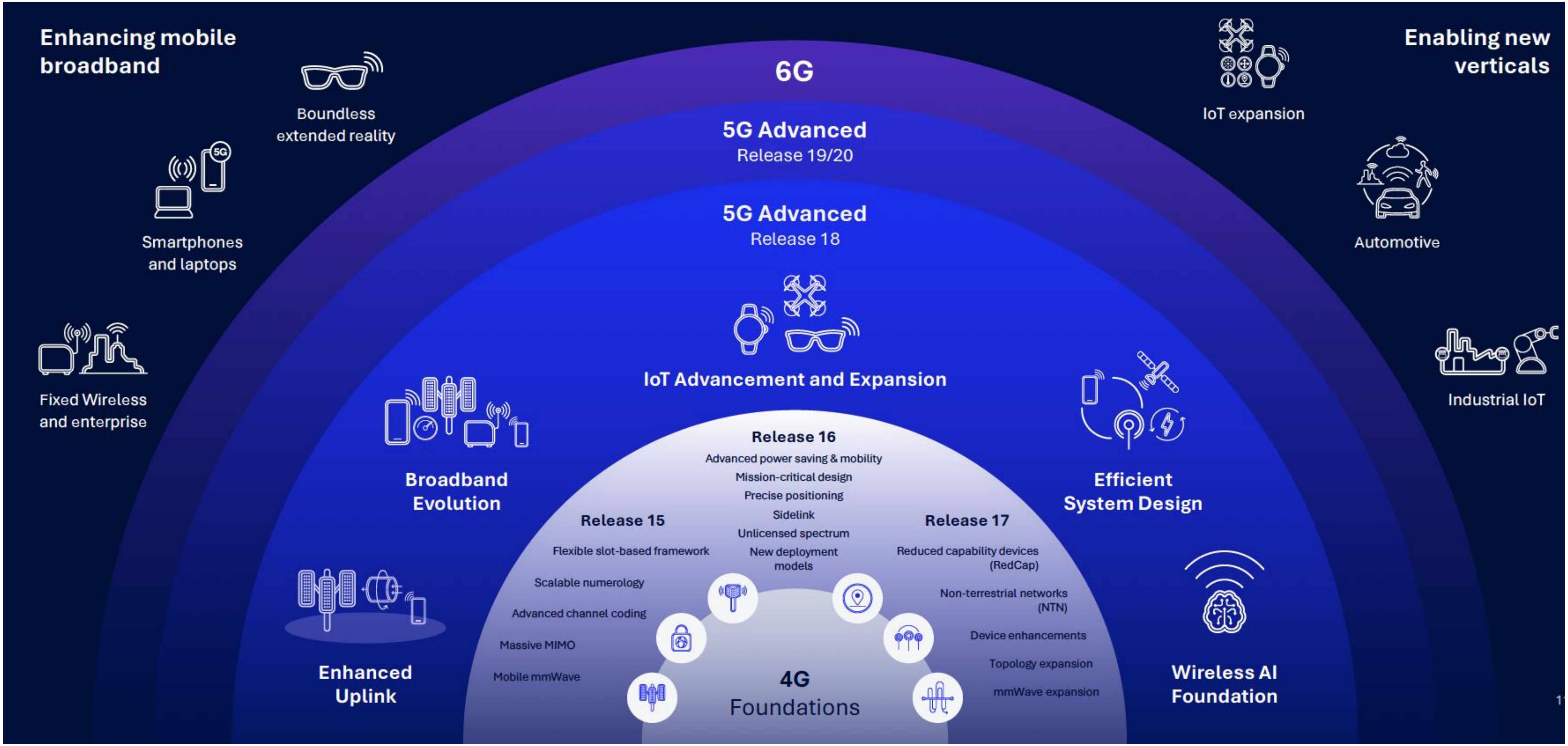
- Release 15 — 5G foundation
  - 5G New Radio introduction
  - Initial massive IoT support
  - Early ultra-reliable low-latency features
  - Edge computing architectural support
- Release 16 — Reliability and industrial enhancement
  - Improved URLLC mechanisms
  - Time-sensitive networking support
  - Industrial IoT integration improvements
  - Better deterministic communication support

Releases relevant to IoT and NTN

- Release 17 — First NTN integration milestone
  - Non-terrestrial network support introduced
  - Satellite access for NR
  - NB-IoT over satellite profiles
  - Direct IoT-to-satellite scenarios enabled
  - NTN device and link adaptations defined
- Release 18 — 5G Advanced phase
  - Enhanced NTN capabilities
  - AI and ML for radio optimisation
  - Reduced capability device class for IoT
  - Energy efficiency improvements
  - Better uplink efficiency for IoT traffic

Releases relevant to IoT and NTN

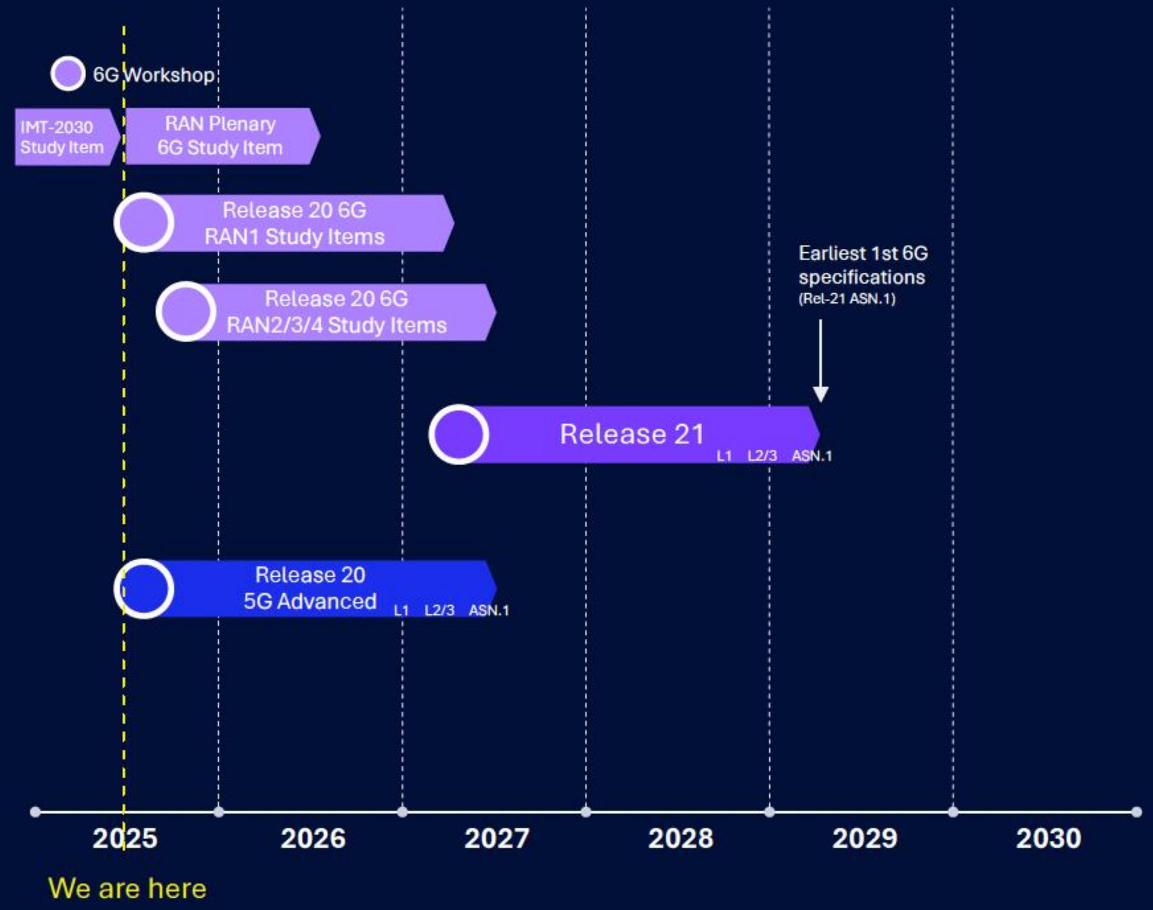
- Release 19 and beyond — Bridge toward 6G
  - Expanded NTN features
  - AI-native radio control studies
  - Integrated sensing and communication studies
  - Pre-6G candidate technology exploration
- Standards pathway insight
  - IMT-2030 defines destination
  - 3GPP Releases define stepping stones
- Checkpoint question
  - Which Release first enabled practical satellite IoT within cellular standards, and why is that important?



Source: Qualcomm

3GPP Release 20 to begin now and expected to complete by June 2027*

A closer look at the Release 20 timeline



* Now = June 2025, ASN.1 milestone expected in June 2027

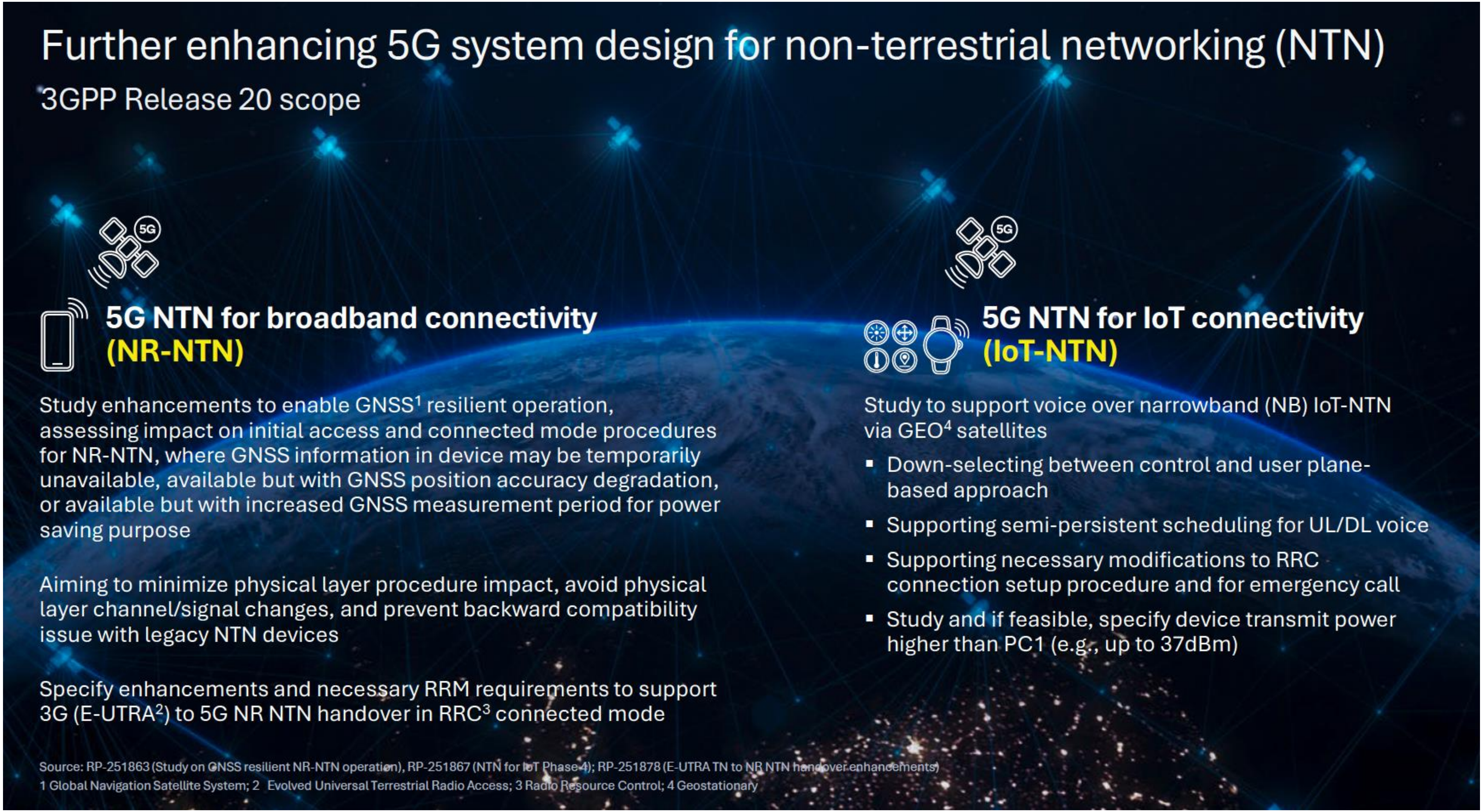
Source: Qualcomm

Laying the foundation for next-generation wireless system

- First release with 6G Study Items
- Focusing on the foundational aspects of the wireless system
- Sufficient time to be allocated to 6G study

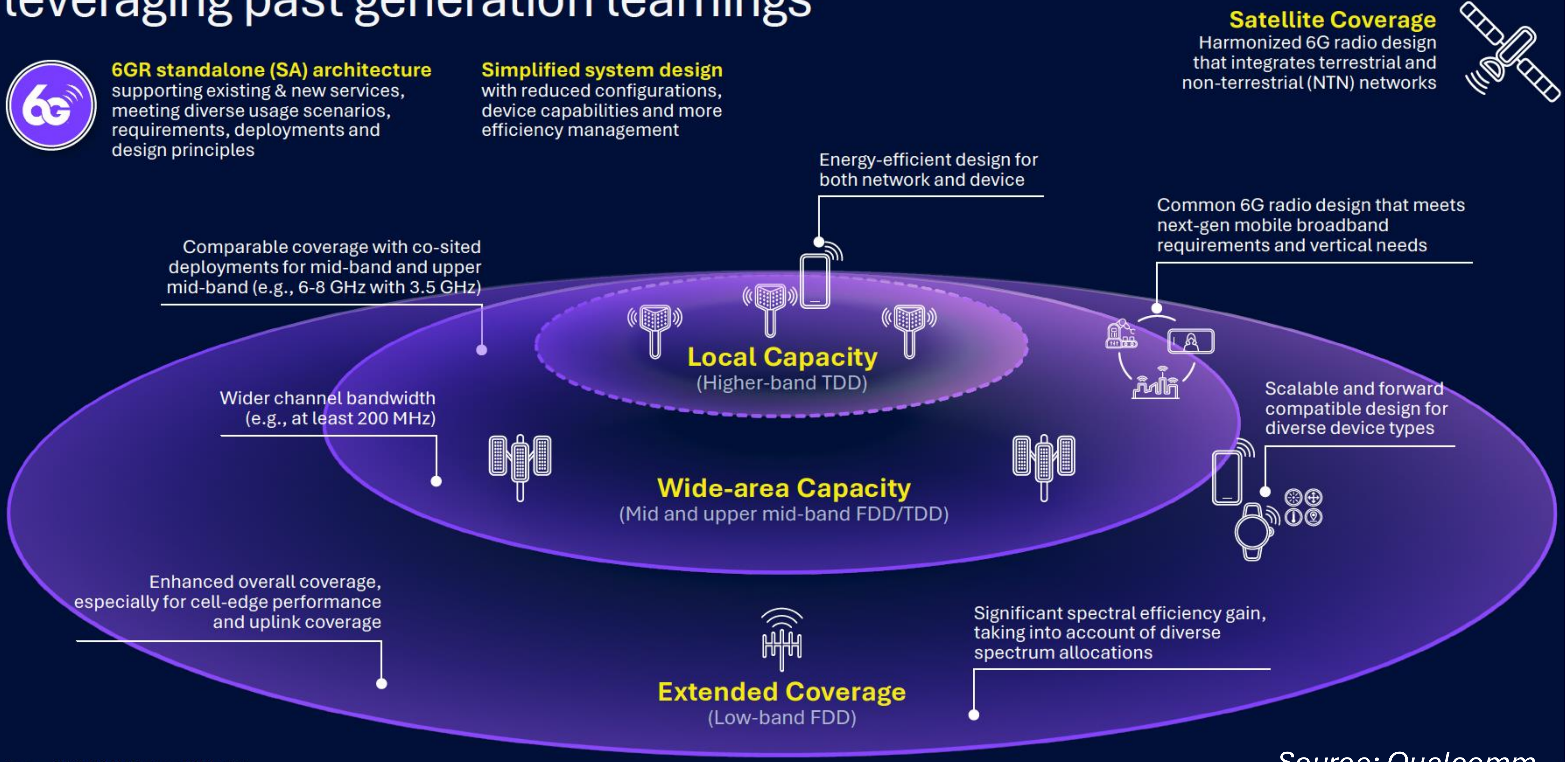
Completing the 5G Advanced evolution for its full potential

- Last release with 5G-only projects (6th release of 5G, 3rd release of 5G Advanced)
- Reasonable time to be allocated for 5G Advanced
- No approval of any 5G Advanced Study Items targeting the entire release duration
- Must consider commercial uptake of previous releases and focus on critical needs only



Source: Qualcomm

Establishing a unified technology framework leveraging past generation learnings



Source: RP-251881 (Study on 6G Radio)

Source: Qualcomm

Long-Range and Integrated NTN IoT Architecture

Motivation for NTN IoT

Motivation for NTN IoT

Remote and rural IoT coverage gaps

Maritime and aerial IoT systems

Disaster and emergency connectivity

Global sensor infrastructures

Long-range IoT application

Smart agriculture at regional scale

Maritime logistics monitoring

Oil, gas, and energy infrastructure sensing

Environmental and climate sensor grids

Disaster monitoring networks

Network layers, payload models

NTN network layers

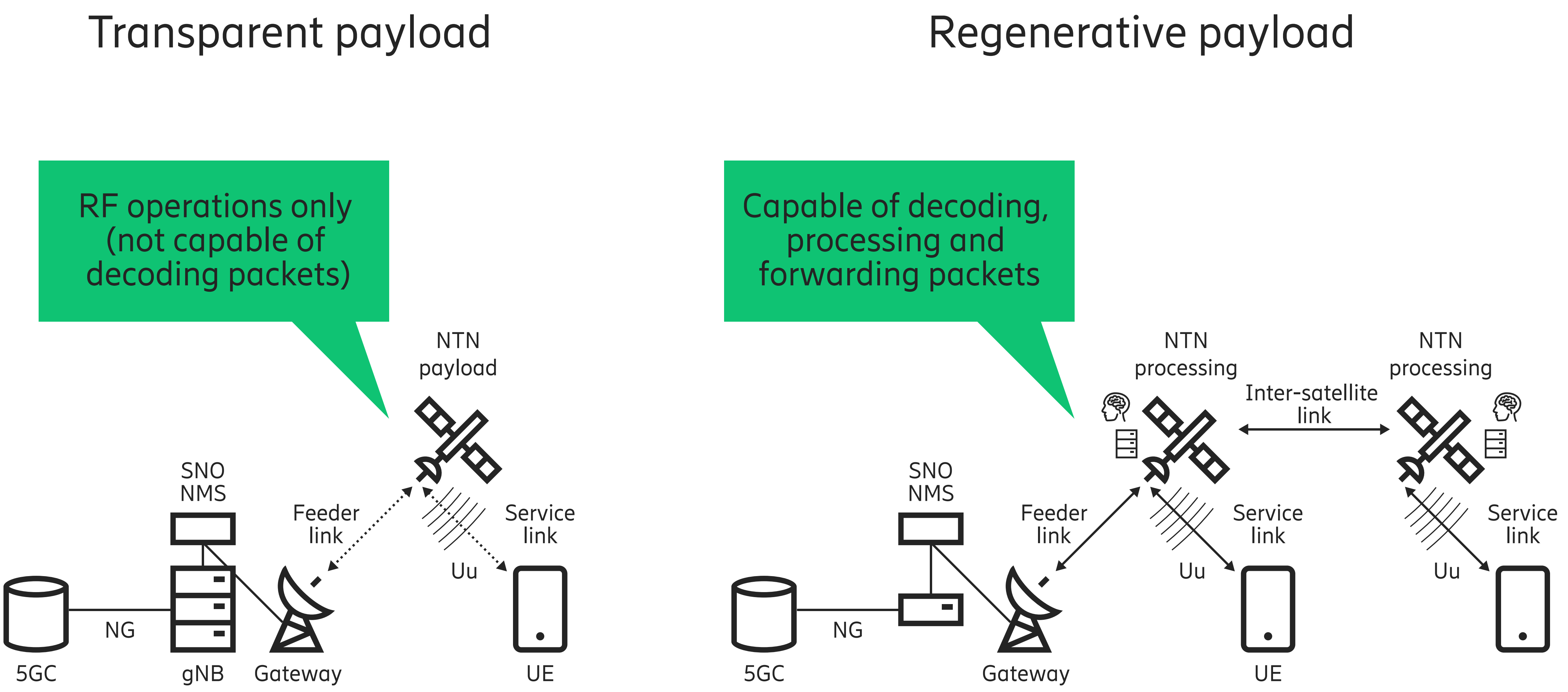
- LEO satellite constellations
- High-altitude platform stations
- UAV base stations
- Terrestrial macro and small cells

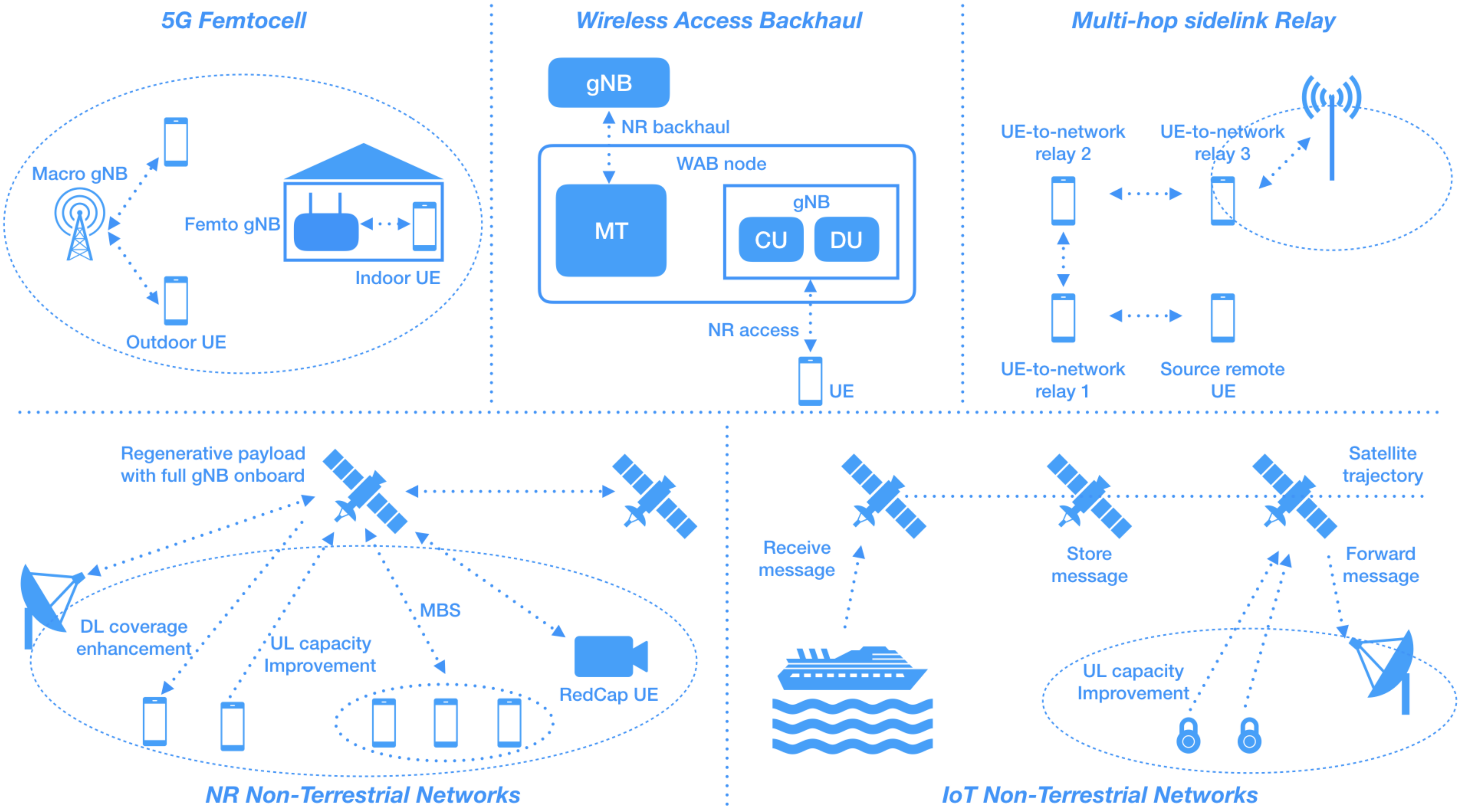
NTN payload models

- Transparent payload relay
- Regenerative payload processing
- Satellite as relay vs satellite as base station

3GPP NTN architectural concepts

- NTN-enabled base stations
- Satellite access adaptation layers
- NTN-aware radio profiles
- IoT NTN device categories

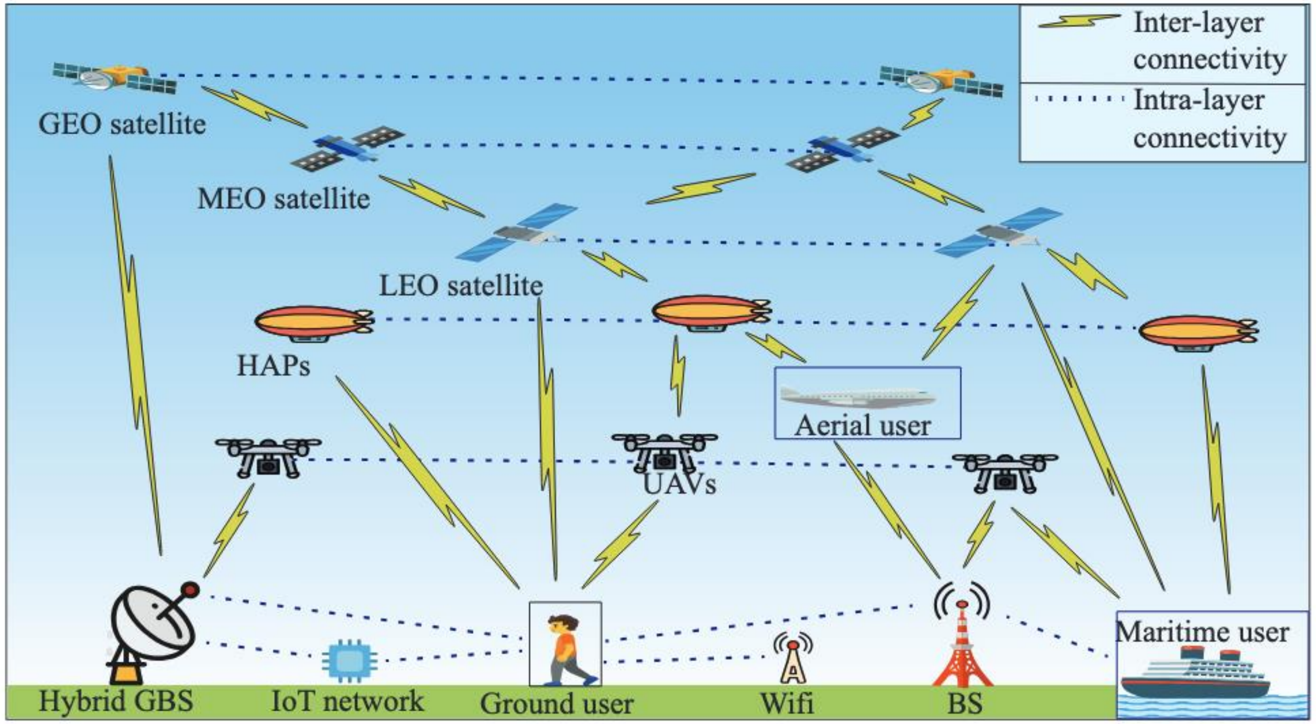

Transparent payload
RF operations only (not capable of decoding packets)
NTN payload
SNO NMS
Feeder link
Service link
Uu
NG
5GC
gNB
Gateway
UE
Regenerative payload
Capable of decoding, processing and forwarding packets
NTN processing
Inter-satellite link
NTN processing
SNO NMS
Feeder link
Service link
Uu
Service link
Uu
NG
5GC
Gateway
UE



An illustration of the key 3GPP Release-19 features related to further evolved 5G topology [1].

Connectivity Model

- Multi-layer connectivity model
  - Terrestrial + satellite dual connectivity
  - UAV-assisted relay paths
  - Path diversity for reliability
- Mobility and session continuity
  - Cross-domain handover
  - Delay-aware mobility control
  - Service continuity across layers



AI-Native Networking and Slicing for Wide-Area IoT

Network slicing within network domains

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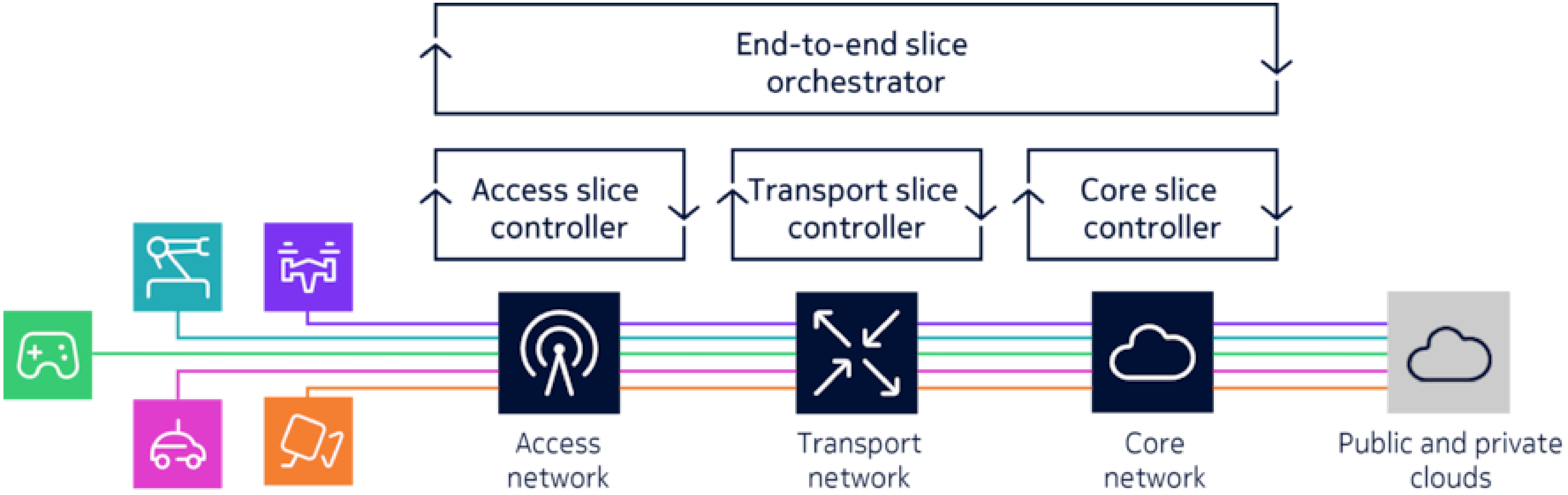
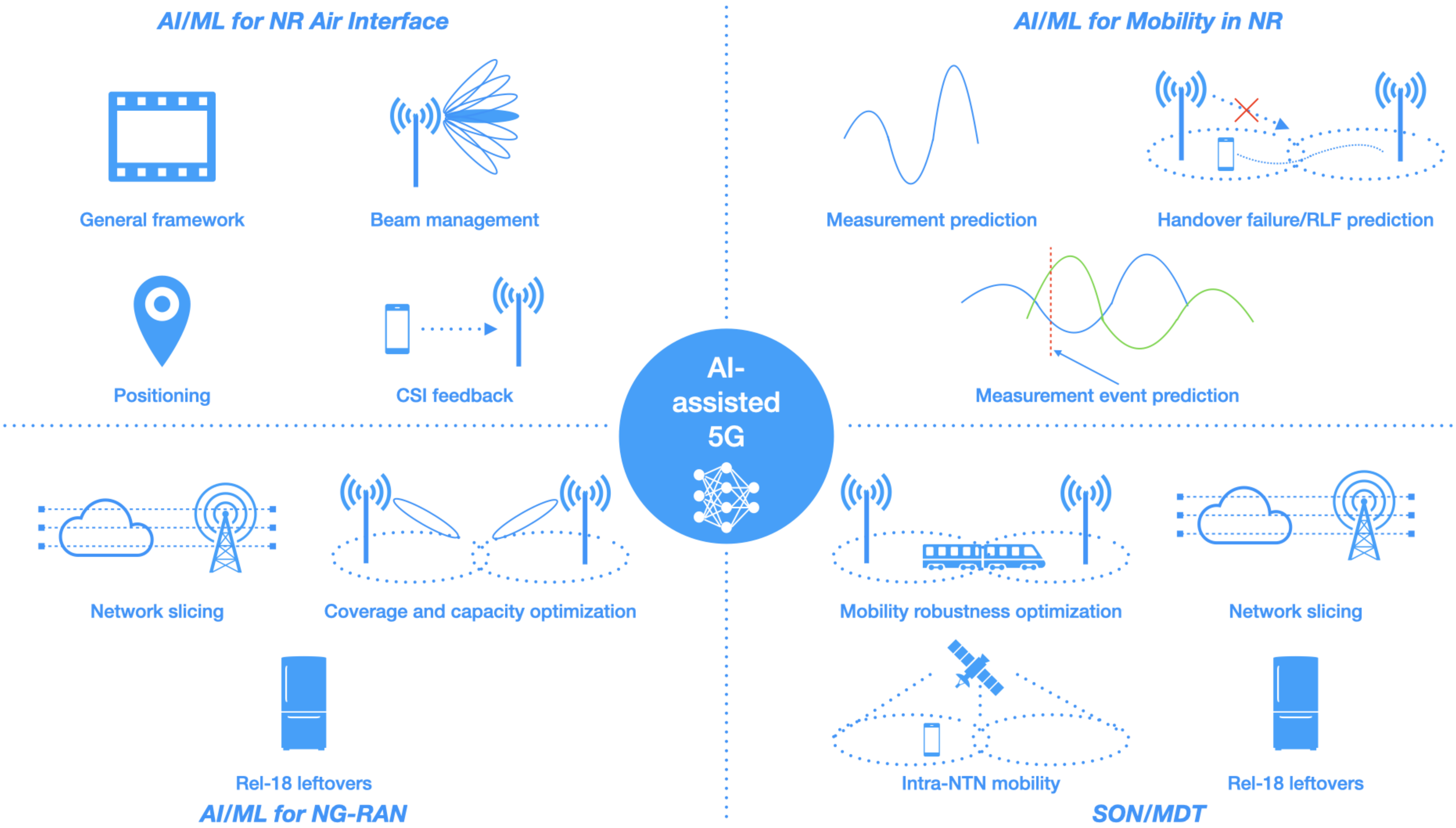


Source: Nokia

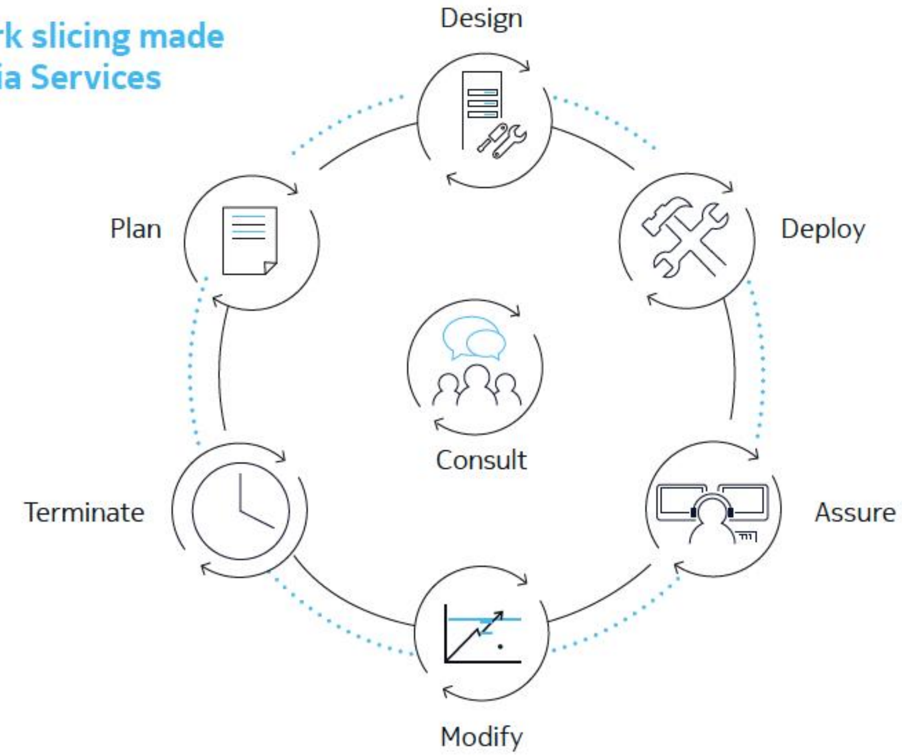


An illustration of the key 3GPP Release-19 features related to AI-assisted 5G [1].

AI-Native Networking for Wide-Area IoT

- AI-native network control
  - Traffic prediction for IoT loads
  - Satellite resource scheduling
  - Congestion and interference prediction
- Closed-loop optimisation
  - Telemetry collection
  - Learning and prediction models
  - Automatic parameter adjustment

Figure 2. Network slicing made simple with Nokia Services



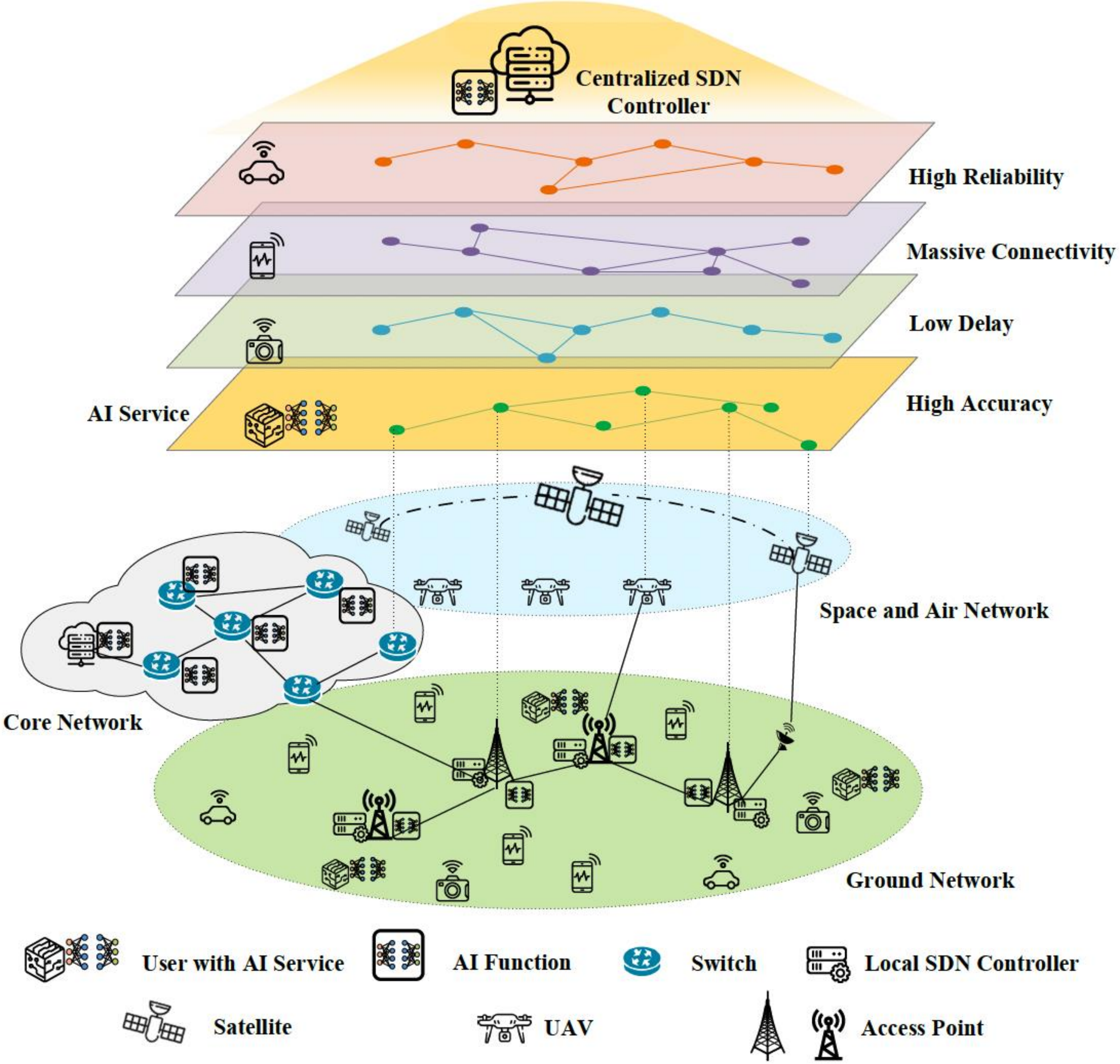
IoT-oriented network slicing

- IoT-oriented network slicing
  - Massive IoT slice — scale and efficiency
  - Critical IoT slice — reliability and latency
  - Global NTN IoT slice — coverage priority
- Cross-domain slice orchestration
  - Terrestrial and satellite joint slice control
  - Policy-driven resource allocation

Activity: Design three slices for global smart agriculture IoT

AI-native network slicing architecture for 6G IoT [2]

- Space-air-ground integrated networks (SAGIN)
- Diversified services
- Ubiquitous intelligence



Deployment Challenges and Testbeds (NTN Focus)

Deployment Challenges and Testbeds

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Measurement metrics

- Latency distribution
- Reliability probability
- Energy per transmitted bit
- Handover success rate

Deployment challenges

- Satellite link budget limits
- Doppler and fast mobility
- Terminal antenna constraints
- Device energy limits

Cost and scalability issues

- Device cost vs coverage benefit
- Infrastructure investment trade-offs
- Shared satellite resource limits

Deployment Challenges and Testbeds

Interoperability and migration

- Coexistence with 5G IoT networks
- Release 17/18 device compatibility
- Gradual upgrade pathways

Testbed directions

- NTN network simulators
- Hybrid satellite-terrestrial labs
- Delay-injection emulation platforms

Summary and Looking Ahead

- Key takeaways
  - IMT-2030 defines 6G IoT capability targets
  - 3GPP Releases define implementation pathway
  - NTN integration is central for long-range IoT
  - AI-native control enables scalable IoT operations
- Design principles
  - Multi-layer connectivity
  - Slice-based service isolation
  - Edge-assisted processing
  - Standards-aligned deployment
- Final reflection question
  - Which IoT domain benefits most from NTN-enabled 6G, and why?
- Week 6: IoT Security!

References

[1] X. Lin, "The Bridge Toward 6G: 5G-Advanced Evolution in 3GPP Release I9," in *IEEE Communications Standards Magazine*, vol. 9, no. 1, pp. 28-35, March 2025, doi: 10.1109/MCOMSTD.0001.2300063.

[2] W. Wu et al., "AI-Native Network Slicing for 6G Networks," in IEEE Wireless Communications, vol. 29, no. 1, pp. 96-103, February 2022, doi: 10.1109/MWC.001.2100338.

[3] Recommendation ITU-R M.2160-0 (11/2023)